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IC socket with high speed contacts

Abstract

An electrical connector having a housing with a first surface and a second surface. Contact receiving openings extend from the first surface to the second surface. Contacts are positioned in the contact receiving openings. The contacts have base members and resilient contact beams extending from opposed sides of the base members. Sidewalls extend from opposed sides of the base members from which the resilient contact beams do not extend. The contacts also provide multiple electrical pathways across the contacts for electrical signal transmissions, allowing for signal transmissions to be transmitted at frequencies of approximately 224 Gbps or greater.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to contact and an electrical connector for electrically connecting an integrated circuit (IC) to a circuit board. In particular, the invention is directed to a contact which allows for high speed signal transmission while providing sufficient normal force to establish a proper electrical connection.

BACKGROUND OF THE INVENTION

(2) Integrated circuits have a number of contact pads which are electrically connected to a circuit board through a connector. Contacts of a number equal to the number of the pads are correspondingly arrayed on an IC socket. Such contacts and sockets are known in the industry.

(3) Recently, ICs have begun processing high-speed signals such as a signal at 100 GHz or above. The current IC sockets and contacts generally cannot accommodate such signal speeds.

(4) Consequently, there is a need for IC sockets which have contacts that can transmit such high-speed signals while also providing the normal forces required to make a proper electrical connection.

SUMMARY OF THE INVENTION

(5) An embodiment is directed to an electrical connector having a housing with a first surface and a second surface. Contact receiving openings extend from the first surface to the second surface. Contacts are positioned in the contact receiving openings. The contacts have base members and resilient contact beams extending from opposed sides of the base members. Sidewalls extend from

opposed sides of the base members from which the resilient contact beams do not extend.

(6) The connector and contacts is configured to provide sufficient normal force to ensure that a positive electrical and mechanical connection is made between the contacts and mating components, such as integrated circuits and circuit boards. The contacts also provide multiple electrical pathways across the contacts for electrical signal transmissions, allowing for signal transmissions to be transmitted at frequencies of approximately 224 Gbps or greater.

(7) Other features and advantages of the present invention will be apparent from the following more detailed description of the preferred embodiment, taken in conjunction with the accompanying drawings which illustrate, by way of example, the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of an illustrative electrical connector assembly with contacts according to the present invention for electrically connecting an integrated circuit to a circuit board.

(2) FIG. 2 is a top view of the electrical connector of FIG. 1.

(3) FIG. 3 is a side view of the electrical connector of FIG. 1.

(4) FIG. 4 is a cross sectional view taken along line 4-4 of FIG. 1.

(5) FIG. 5 is a cross sectional view similar to FIG. 4 with the illustrative connector mounted between an integrated circuit and circuit board.

(6) FIG. 6 is a perspective view of an illustrative contact of the connector of FIG. 1.

(7) FIG. 7 is a side view of the contact of FIG. 6.

(8) FIG. 8 is a side view of the contact of FIG. 6.

(9) FIG. 9 is a perspective view of another illustrative electrical connector with contacts according to the present invention for electrically connecting an integrated circuit to a circuit board, the contacts are shown in the closed or stressed position.

(10) FIG. 10 is a top view of the electrical connector of FIG. 9.

(11) FIG. 11 is a side view of a representative closed contact of the electrical connector of FIG. 9.

(12) FIG. 12 is a side view of a representative open contact of the electrical connector of FIG. 9.

(13) FIG. 13 is a top perspective view of an illustrative contact of the connector of FIG. 9.

(14) FIG. 14 is a bottom perspective view of an illustrative contact of the connector of FIG. 13.

(15) FIG. 15 is a top view of the contact of FIG. 13.

DETAILED DESCRIPTION OF THE INVENTION

(16) The description of illustrative embodiments according to principles of the present invention is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. In the description of embodiments of the invention disclosed herein, any reference to direction or orientation is merely intended for convenience of description and is not intended in any way to limit the scope of the present invention. Relative terms such as “lower,” “upper,” “horizontal,” “vertical,” “above,” “below,” “up,” “down,” “top” and “bottom” as well as derivative thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description only and do not require that the apparatus be constructed or operated in a particular orientation unless explicitly indicated as such. Terms such as “attached,” “affixed,” “connected,” “coupled,” “interconnected,” and similar refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise.

(17) Moreover, the features and benefits of the invention are illustrated by reference to the preferred embodiments. Accordingly, the invention expressly should not be limited to such

embodiments illustrating some possible non-limiting combination of features that may exist alone or in other combinations of features, the scope of the invention being defined by the claims appended hereto.

(18) As shown in FIGS. 1-4, a first illustrative electrical connector assembly **10** has a housing **12** with a top or first surface **14** and an oppositely facing bottom or second surface **16**. Sidewalls **18** extend between the top surface **14** and the bottom surface **16**. In the illustrative embodiment shown, the housing has a generally rectangular shape, but other configurations of the housing may be used. The housing **12** is made from a nonconductive material which has the mechanical strength required.

(19) Contact receiving openings **20** are provided in the housing **12**. The contact receiving openings **20** extend from the top surface **14** to the bottom surface **16**. The number of the contact receiving openings **20** may vary depending on the number of electrical connections required between an integrated circuit **17** (FIG. 5) positioned on the top surface **14** and a circuit board **19** (FIG. 5) positioned on the bottom surface **16**. The contact receiving openings **20** have first sidewalls **22** and second sidewalls **24**. Each contact receiving opening **20** has two opposed first sidewalls **22** and two opposed second sidewalls **24**.

(20) Contacts **30** are positioned in each of the contact receiving openings **20**. Each of the contacts **30** is formed by stamping and forming a conductive material, such as, but not limited to, a metal plate.

(21) As shown in FIGS. 5-7, the contacts **30** have base members **32**. Resilient contact beams **34** extend from opposed sides of the base members **32**. Sidewalls **35** extend from opposed sides of the base members **32** from which the resilient contact beams **34** do not extend.

(22) In the illustrative embodiment shown, each contact **30** has two primary resilient contact beams **34** which are essentially mirror images of each other. The primary resilient contact beams **34** are bent from the base members **32** and extend obliquely therefrom. The primary resilient contact beams **34** have fixed end portions **36**, arcuate or curved portions **38** and contact portions **40**. The fixed end portions **36** extend from and are integrally attached to the base members **32**. The arcuate or curved portions **38** extend from the fixed end portions **36** to the contact portions **40**. The contact portions **40** extend from the arcuate or curved portions **38** and are provided at the free ends of the primary resilient contact beams **34**.

(23) The contact portions **40** have contact areas **42** and secondary resilient beams **44**. In the embodiment shown, the contact areas **42** have a curved, convex shape and are positioned proximate the free ends of the primary resilient contact beams **34**. The contact areas **42** are provided in line with the longitudinal axes of the primary resilient contact beams **34**. The secondary resilient beams **44** extend from either side of the primary resilient contact beams **34**. The secondary resilient beams **44** are bent such that free ends **46** of the secondary resilient beams **44** extend outside the plane of the contact areas **42**. The contact area **42** and the secondary resilient beams **44** of each contact **30** form a U-shaped member as viewed in FIG. 7.

(24) The sidewalls **35** are planar members connected to the base members **32** by linking members **48** which extend between the sidewalls **35** and the base members **32**. Each contact **30** has two linking members **48** which extend from opposite sides of the base member **32**. The linking members **48** extend from the sides of the base member **32** from which no primary resilient contact beam **34** extends. The linking members **48** are bent such that the planes of the sidewalls **35** extend in a direction which is essentially perpendicular to the planes of the base members **32**.

(25) The sidewalls **35** have inside surfaces **50** and outside surfaces **52**. Retention projections **54** extend from the sidewalls **35**. In the illustrative embodiment shown, the retention projections **54** are rounded and project from a side surfaces of the sidewalls **35**. However, other configurations of the retention projections **54** may be used.

(26) As shown in FIGS. 6 through 8, when each contact **30** is formed, the free ends **46** of the secondary resilient beams **44** are provided in mechanical and electrical engagement with inside

surfaces **50** of the sidewalls **35**. The free ends **46** are configured to move relative to the inside surfaces **50** as force is applied to the contact areas **42** of the contacts **30**, as will be more fully described.

(27) As shown in FIGS. **1** through **5**, the contacts **30** are positioned in the contact receiving openings **20** of the housing **12** of the electrical connector assembly **10**. In this position, the base members **32** of the contacts **30** are positioned proximate to or in engagement with respective first sidewalls **22** of the contact receiving openings **20**. In addition, the sidewalls **35** of the contacts **30** are positioned proximate to or in engagement with the second sidewalls **24** of the contact receiving openings **20**. The retention projections **54** of the sidewalls **35** of the contacts **30** engage respective first sidewalls **22** of the contact receiving openings **20**. The interaction and the engagement of the base members **32**, the sidewalls **35** and the retention projections **54** of the contacts **30** with the first sidewalls **22** and the second sidewalls **24** of the contact receiving openings **20** provides an interference fit or press fit between the contacts **30** and the contact receiving openings **20** to mechanically secure the contacts **30** in the contact receiving openings **20**.

(28) As shown in FIGS. **6** through **8**, when each contact **30** is formed, the free ends **46** of the secondary resilient beams **44** are provided in mechanical and electrical engagement with inside surfaces **50** of the sidewalls **35**. The free ends **46** are configured to move relative to the inside surfaces **50** as force is applied to the contact areas **42** of the contacts **30**, as will be more fully described.

(29) With contacts **30** properly secured in the contact receiving openings **20**, the contact portions **40** of respective primary resilient contact beams **34** extends or protrudes beyond the top surface **14** of the housing **12** so that conductive pads (not shown in the figures) of the integrated circuit **17** electrically and mechanically contacts the contact portions **40**. Similarly, the contact portions **40** of other respective primary resilient contact beams **34** extends or protrudes beyond the bottom surface **16** of the housing **12** so that conductive pads (not shown in the figures) of the circuit board **19** electrically and mechanically contacts the contact portions **40**.

(30) In operation, the electrical connector assembly **10** is moved into position relative the circuit board **19**, such that respective bottom contact portions **40** of the contacts **30** are provided in line with the pads of the circuit board **19**. The housing **12** of the connector assembly **10** is moved into engagement with the pads of the circuit board **19**. As this occurs, force is applied to the bottom contact portions **40** which extend from the bottom surface **16** of the housing **12**. This causes the bottom contact portions **40** to move in a direction toward the top surface **14** of the housing **12**. As this occurs, the respective bottom primary resilient contact beams **34** are elastically deformed. In addition, as the bottom contact portions **40** are moved, the free ends **46** of the bottom secondary resilient beams **44** are frictionally slide across the inside surfaces **50** of the sidewalls **35**. The elastic deformation of the bottom primary resilient contact beams **34** and the frictional engagement of the free ends **46** of the bottom secondary resilient beams **44** with the inside surfaces **50** of the sidewalls **35** as the free ends **46** are moved cause the bottom contact portions **40** to exert a normal force on the pads of the circuit board **19**. This allows the bottom contact portions **40** to be retained in mechanical and electrical engagement with the contact pads of the circuit board **19**. The movement of the bottom contact portions **40** is greater than approximately 0.05 mm, approximately 0.1 mm, less than approximately 0.5 mm. The normal force exerted by the bottom contact portions **40** is greater than approximately 15 grams, greater than approximately 20 grams, between approximately 15 grams and 30 grams, between approximately 20 grams and 25 grams. The connector assembly **10** is secured to the circuit board **19** using known methods.

(31) An integrated circuit **17** is moved into position relative to the electrical connector assembly **10**, such that respective top contact portions **40** of the contacts **30** are provided in line with the pads of the integrated circuit **17**. The integrated circuit **17** is moved into engagement with housing **12** of the connector assembly **10**. As this occurs, force is applied to the top contact portions **40** which extend from the top surface **14** of the housing **12**. This causes the top contact portions **40** to move in a

direction toward the bottom surface **16** of the housing **12**. As this occurs, the respective top primary resilient contact beams **34** are elastically deformed. In addition, as the top contact portions **40** are moved, the free ends **46** of the top secondary resilient beams **44** are frictionally slide across the inside surfaces **50** of the sidewalls **35**. The elastic deformation of the top primary resilient contact beams **34** and the frictional engagement of the free ends **46** of the top secondary resilient beams **44** with the inside surfaces **50** of the sidewalls **35** as the free ends **46** are moved cause the top contact portions **40** to exert a normal force on the pads of the integrated circuit **17**. This allows the top contact portions **40** to be retained in mechanical and electrical engagement with the contact pads of the integrated circuit **17**. The movement of the bottom contact portions **40** is greater than approximately 0.05 mm, approximately 0.1 mm, less than approximately 0.5 mm. The normal force exerted by the top contact portions **40** is greater than approximately 15 grams, greater than approximately 20 grams, between approximately 15 grams and 30 grams, between approximately 20 grams and 25 grams. The integrated circuit **17** is secured to the connector assembly **10** using known methods.

(32) With the connector assembly **10** properly connected to the integrated circuit **17** and the circuit board **19**, each contact **30** provides multiple electrical pathways for electrical signal transmissions. The first electrical pathway, as shown by arrow A in FIG. 7, is between the integrated circuit pad, the top contact area **42** of the top contact portion **40**, the top arcuate or curved portion **38**, the top fixed end portion **36**, the base member **32**, the bottom fixed end portion **36**, the bottom arcuate or curved portion **38**, the bottom contact area **42** of the bottom contact portion **40** and the circuit board pad. The second electrical pathway, as shown by arrow B in FIG. 6, is between the integrated circuit pad, the top contact area **42** of the top contact portion **40**, the two top secondary resilient beams **44**, the sidewalls **35**, the two bottom secondary resilient beams **44**, the bottom contact area **42** of the bottom contact portion **40** and the circuit board pad. The allows the signal transmissions to transmitted at frequencies of approximately 224 Gbps or greater.

(33) As shown in FIGS. 9-11, a second illustrative electrical connector assembly **110** has a housing **112** with a top surface **114** and an oppositely facing bottom surface **116**. Sidewalls **118** extend between the top surface **114** and the bottom surface **116**. In the illustrative embodiment shown, a partial view of the housing **112**. The housing may have a generally rectangular shape, but other configurations of the housing may be used. The housing **112** is made from a nonconductive material which has the mechanical strength required.

(34) Contact receiving openings **120** are provided in the housing **112**. The contact receiving openings **120** extend from the top surface **114** to the bottom surface **116**. The number of the contact receiving openings **120** may vary depending on the number of electrical connections required between an integrated circuit **117** (FIG. 11) positioned on the top surface **114** and a circuit board **19** (FIG. 11) positioned on the bottom surface **116**. The contact receiving openings **120** have first sidewalls **122** and second sidewalls **124**. Each contact receiving opening **120** has two opposed first sidewalls **122** and two opposed second sidewalls **124**. Respective sidewalls **122** many be contoured to accommodate the shape of the contacts **130**.

(35) Contacts **130** are positioned in each of the contact receiving openings **120**. Each of the contacts **130** is formed by stamping and forming a conductive material, such as, but not limited to, a metal plate.

(36) As shown in FIGS. 13-15, the contacts **130** have base members **132**. The base members **132** have angled sections **133** which extend from each side of the base members **132**. Resilient contact beams **134** extend from opposed sides of the angled sections **133** of the base members **32**. In the embodiment shown, each contact **130** has four resilient contact beams **134**. Sidewalls **135** extend from the ends of the angled section **133** of the base members **132**.

(37) In the illustrative embodiment shown, each contact **30** has four primary resilient contact beams **134**, with two extending from each angled section **133**. The two which extend from same angled section **133** are essentially mirror images of each other. The primary resilient contact beams **134** are

bent from the angled sections **133** of the base members **132** and extend obliquely therefrom. The primary resilient contact beams **134** have fixed end portions **136**, arcuate or curved portions **138** and contact portions **140**. The fixed end portions **136** extend from and are integrally attached to the angled sections **133** of the base members **132**. The arcuate or curved portions **138** extend from the fixed end portions **136** to the contact portions **140**. The contact portions **140** extend from the arcuate or curved portions **138** and are provided at the free ends of the primary resilient contact beams **134**.

(38) The contact portions **140** have contact areas **142** and secondary resilient beams **144**. In the embodiment shown, the contact areas **142** have a curved, convex shape and are positioned proximate the free ends of the primary resilient contact beams **134**. The contact areas **142** are provided in line with the longitudinal axes of the primary resilient contact beams **134**. The secondary resilient beams **144** extend from one side of the primary resilient contact beams **134**. The side from which the secondary resilient beams **144** extend varies depending upon the positioning of the primary resilient contact beams **134**. The secondary resilient beams **144** are bent such that free ends **146** of the secondary resilient beams **144** extend outside the plane of the contact areas **142**. The contact area **142** and the secondary resilient beam **144** of each primary resilient contact beams **134** of a half of a U-shaped member, while two adjacent contact areas **142** and secondary resilient beams **144** form a generally U-shaped member as viewed in FIG. **13**.

(39) The sidewalls **135** are planar members connected to the angled sections **133** of the base members **132** by linking members **148** which extend between the sidewalls **135** and the angled sections **133** of the base members **132**. Each contact **130** has two linking members **148** which extend from opposite sides of the base member **132**. The linking members **148** extend from the ends of the angled sections **133** of the base member **32** from which no primary resilient contact beam **134** extends. The linking members **148** are bent such that the planes of the sidewalls **135** extend in a direction which approached, but is less than, perpendicular to the planes of the base members **132**. The sidewalls **135** have inside surfaces **150** and outside surfaces **152**. As shown in FIGS. **13** through **15**, when each contact **130** is formed, the free ends **146** of the secondary resilient beams **144** are spaced from the inside surfaces **150** of the sidewalls **135**.

(40) As shown in FIGS. **9** through **12**, the contacts **130** are positioned in the contact receiving openings **120** of the housing **112** of the electrical connector assembly **110**. In this position, the base members **132**, including the angled sections **133** of the contacts **130** are positioned proximate to or in engagement with respective first sidewalls **122** of the contact receiving openings **120**. In addition, the sidewalls **135** of the contacts **130** are positioned proximate to or in engagement with the second sidewalls **124** of the contact receiving openings **120**. The positioning of the sidewalls **135** in engagement with the second sidewalls **124** of the contact receiving openings **120** causes the linking member **148** to be elastically deformed and the sidewalls **135** of each contact **130** to be moved toward each other.

(41) The interaction and the engagement of the base members **132** and the sidewalls **135** with the first sidewalls **122** and the second sidewalls **124** of the contact receiving openings **120** provides an interference fit or press fit between the contacts **130** and the contact receiving openings **120** to mechanically secure the contacts **130** in the contact receiving openings **120**.

(42) As shown in FIGS. **9** and **10**, when each contact **130** is properly positioned in the contact receiving openings **120**, the free ends **146** of the secondary resilient beams **144** are provided in mechanical and electrical engagement with inside surfaces **150** of the sidewalls **135**. The free ends **146** are configured to move relative to the inside surfaces **150** as force is applied to the contact areas **142** of the contacts **130**, as will be more fully described.

(43) As shown in FIG. **12**, with contacts **130** properly secured in the contact receiving openings **120**, the contact portions **140** of respective primary resilient contact beams **134** extends or protrudes beyond the top surface **114** of the housing **112** so that conductive pads (not shown in the figures) of the integrated circuit **117** electrically and mechanically contacts the contact portions

140. Similarly, the contact portions **140** of other respective primary resilient contact beams **134** extends or protrudes beyond the bottom surface **116** of the housing **112** so that conductive pads (not shown in the figures) of the circuit board **119** electrically and mechanically contacts the contact portions **140**.

(44) As shown in FIG. **11**, in operation, the electrical connector assembly **110** is moved into position relative the circuit board **119**, such that respective bottom contact portions **140** of the contacts **130** are provided in line with the pads of the circuit board **119**. The housing **112** of the connector assembly **110** is moved into engagement with the pads of the circuit board **119**. As this occurs, force is applied to the bottom contact portions **140** which extend from the bottom surface **116** of the housing **112**. This causes the bottom contact portions **140** to move in a direction toward the top surface **114** of the housing **112**. As this occurs, the respective bottom primary resilient contact beams **134** are elastically deformed. In addition, as the bottom contact portions **140** are moved, the free ends **146** of the bottom secondary resilient beams **144** are frictionally slide across the inside surfaces **150** of the sidewalls **135**. The elastic deformation of the bottom primary resilient contact beams **134** and the frictional engagement of the free ends **146** of the bottom secondary resilient beams **144** with the inside surfaces **150** of the sidewalls **135** as the free ends **146** are moved cause the bottom contact portions **140** to exert a normal force on the pads of the circuit board **119**. This allows the bottom contact portions **140** to be retained in mechanical and electrical engagement with the contact pads of the circuit board **119**. The movement of the bottom contact portions **140** is greater than approximately 0.8 mm, great than approximately 0.9 mm, greater than approximately 1.0 mm, less than approximately 1.5 mm. The normal force exerted by the bottom contact portions **140** is greater than approximately 15 grams, greater than approximately 20 grams, between approximately 15 grams and 30 grams, between approximately 20 grams and 25 grams. The connector assembly **110** is secured to the circuit board **119** using known methods.

(45) An integrated circuit **117** is moved into position relative to the electrical connector assembly **110**, such that respective top contact portions **140** of the contacts **130** are provided in line with the pads of the integrated circuit **117**. The integrated circuit **117** is moved into engagement with housing **112** of the connector assembly **110**. As this occurs, force is applied to the top contact portions **140** which extend from the top surface **114** of the housing **112**. This causes the top contact portions **140** to move in a direction toward the bottom surface **116** of the housing **112**. As this occurs, the respective top primary resilient contact beams **134** are elastically deformed. In addition, as the top contact portions **140** are moved, the free ends **146** of the top secondary resilient beams **144** are frictionally slide across the inside surfaces **150** of the sidewalls **135**. The elastic deformation of the top primary resilient contact beams **134** and the frictional engagement of the free ends **146** of the top secondary resilient beams **144** with the inside surfaces **150** of the sidewalls **135** as the free ends **146** are moved cause the top contact portions **140** to exert a normal force on the pads of the integrated circuit **117**. This allows the top contact portions **140** to be retained in mechanical and electrical engagement with the contact pads of the integrated circuit **117**. The movement of the top contact portions **140** is greater than approximately 0.8 mm, great than approximately 0.9 mm, greater than approximately 1.0 mm, less than approximately 1.5 mm. The normal force exerted by the top contact portions **40** is greater than approximately 15 grams, greater than approximately 20 grams, between approximately 15 grams and 30 grams, between approximately 20 grams and 25 grams. The integrated circuit **117** is secured to the connector assembly **10** using known methods.

(46) With the connector assembly **110** properly connected to the integrated circuit **117** and the circuit board **119**, each contact **130** provides multiple electrical pathways for electrical signal transmissions. The first electrical pathways, as shown by arrow A in FIG. **13**, are between the integrated circuit pads, the top contact areas **142** of the top contact portions **140**, the top arcuate or curved portions **138**, the top fixed end portions **136**, the angled sections **133** of the base member **132**, the bottom fixed end portions **136**, the bottom arcuate or curved portions **138**, the bottom

contact areas **142** of the bottom contact portions **140** and the circuit board pad. The second electrical pathways, as shown by arrow B in FIG. **13**, are between the integrated circuit pad, the top contact areas **142** of the top contact portions **140**, the top secondary resilient beams **144**, the sidewalls **135**, the bottom secondary resilient beams **144**, the bottom contact areas **142** of the bottom contact portions **140** and the circuit board pad. The allows the signal transmissions to transmitted at frequencies of approximately 224 Gbps or greater.

(47) One skilled in the art will appreciate that the invention may be used with many modifications of structure, arrangement, proportions, sizes, materials and components and otherwise used in the practice of the invention, which are particularly adapted to specific environments and operative requirements without departing from the principles of the present invention. The presently disclosed embodiments are therefore to be considered in all respects as illustrative and not restrictive, the scope of the invention being defined by the appended claims, and not limited to the foregoing description or embodiments.

Claims

1. An electrical connector comprising: a housing having a first surface and a second surface, contact receiving openings extending from the first surface to the second surface; contacts positioned in the contact receiving openings, the contacts having base members and primary resilient contact beams extending from opposed sides of the base members, sidewalls extending from opposed sides of the base members from which the primary resilient contact beams do not extend, the primary resilient contact beams having contact portions provided at free ends of the primary resilient contact beams, the contact portions having contact areas and secondary resilient beams; the secondary resilient beams extending from either side of the primary resilient contact beams, the secondary resilient beams are bent such that free ends of the secondary resilient beams extend outside a plane of the contact areas.
2. The electrical connector as recited in claim 1, wherein each contact of the contacts has two primary resilient contact beams which are mirror images of each other.
3. The electrical connector as recited in claim 1, wherein the primary resilient contact beams are bent from the base members and extend obliquely therefrom, the primary resilient contact beams have fixed end portions, arcuate portions and contact portions, the fixed end portions extend from and are integrally attached to the base members, the arcuate portions extend from the fixed end portions to the contact portions.
4. The electrical connector as recited in claim 1, wherein the contact areas have a curved, convex shape and are provided in line with a longitudinal axes of the primary resilient contact beams.
5. The electrical connector as recited in claim 1, wherein the contact area and the secondary resilient beams of each contact form a U-shaped member.
6. The electrical connector as recited in claim 5 **1**, wherein the sidewalls are planar members connected to the base members by linking members which extend between the sidewalls and the base members.
7. The electrical connector as recited in claim 6, wherein the linking members are bent such that planes of the sidewalls extend in a direction which is perpendicular to planes of the base members.
8. The electrical connector as recited in claim 7, wherein the sidewalls have retention projections which extend from the sidewalls.
9. An electrical connector comprising: a housing having a first surface and a second surface, contact receiving openings extending from the first surface to the second surface; contacts positioned in the contact receiving openings, the contacts having base members and primary resilient contact beams extending from opposed sides of the base members, sidewalls extending from opposed sides of the base members from which the primary resilient contact beams do not extend, the primary resilient contact beams having contact portions provided at free ends of the

primary resilient contact beams, the contact portions having contact areas and secondary resilient beams; free ends of the secondary resilient beams provided in mechanical and electrical engagement with inside surfaces of the sidewalls, the free ends being configured to move relative to the inside surfaces as force is applied to the contact areas of the contacts.

10. The electrical connector as recited in claim 9, wherein first contact portions of the contact portions of first respective primary resilient contact beams extend beyond the first surface of the housing and second contact portions of the contact portions of second respective primary resilient contact beams extend beyond the second surface of the housing.

11. The electrical connector as recited in claim 9, wherein each contact provides multiple electrical pathways for electrical signal transmissions, the first electrical pathway is between first contact areas of the first contact portions, the first respective primary resilient contact beam, the base member, the second respective primary resilient contact beam, and second contact areas of the second contact portion, the second electrical pathway, is between the first contact areas of the first contact portion, first secondary beams of the secondary resilient beams, the sidewalls, second secondary beams of the secondary resilient beams, and the second contact areas of the second contact portion.

12. The electrical connector as recited in claim 9, wherein the base members have angled sections from which the resilient contact beams extend.

13. The electrical connector as recited in claim 12, wherein the sidewalls are planar members and are connected to the angled sections of the base members by linking members which extend between the sidewalls and the angled sections of the base members.

14. The electrical connector as recited in claim 13, wherein the linking members are bent such that the planes of the sidewalls extend in a direction which is less than perpendicular to planes of the base members, the sidewalls have inside surfaces, wherein prior to insertion into the contact receiving openings, the free ends of the secondary resilient beams are spaced from the inside surfaces of the sidewalls.

15. The electrical connector as recited in claim 14, wherein when the contacts are positioned in the contact receiving openings of the housing, the angled sections are positioned proximate to or in engagement with first sidewalls of the contact receiving openings and the sidewalls of the contacts are positioned proximate to or in engagement with second sidewalls of the contact receiving openings, the positioning of the sidewalls in engagement with the second sidewalls of the contact receiving openings causes the linking member **14**-to be elastically deformed and the sidewalls of each contact to be moved toward each other.

16. The electrical connector as recited in claim 15, wherein when the contacts are positioned in the contact receiving openings the free ends of the secondary resilient beams are provided in mechanical and electrical engagement with the inside surfaces of the sidewalls, wherein the free ends are configured to move relative to the inside surfaces as force is applied to the contact areas of the contacts.

17. The electrical connector as recited in claim 16, wherein each contact provides multiple electrical pathways for electrical signal transmissions, the first electrical pathway is between first contact areas of the first contact portions, the first respective primary resilient contact beam, the angled sections of the base member, the second respective primary resilient contact beam, and second contact areas of the second contact portion, the second electrical pathway, is between the first contact areas of the first contact portion, first secondary beams of the secondary resilient beams, the sidewalls, second secondary beams of the secondary resilient beams, and the second contact areas of the second contact portion.
